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3次元の極座標表現

$$x = r \sin \theta \cos \phi \quad (1)$$

$$y = r \sin \theta \sin \phi \quad (2)$$

$$z = r \cos \theta \quad (3)$$

(1)(2)(3) より

$$x^2 + y^2 + z^2 = r^2$$

この式の両辺を x で偏微分すると

$$2x = 2r \frac{\partial r}{\partial x}$$

$$\frac{\partial r}{\partial x} = \frac{x}{r} = \sin \theta \cos \phi$$

となる. 同様に

$$y = r \frac{\partial r}{\partial y}, \quad z = r \frac{\partial r}{\partial z}$$

から,

$$\frac{\partial r}{\partial y} = \sin \theta \sin \phi$$

$$\frac{\partial r}{\partial z} = \cos \theta$$

が得られる. 次に (1) (2) より

$$x \sin \phi = y \cos \phi$$

この両辺を x で偏微分すると

$$\sin \phi + x \cos \phi \frac{\partial \phi}{\partial x} = -y \sin \phi \frac{\partial \phi}{\partial x}$$

$$(r \sin \theta \cos \phi) \cos \phi \frac{\partial \phi}{\partial x} + (r \sin \theta \sin \phi) \sin \phi \frac{\partial \phi}{\partial x} = -\sin \phi$$

$$r \sin \theta \frac{\partial \phi}{\partial x} = -\sin \phi$$

$$\frac{\partial \phi}{\partial x} = -\frac{\sin \phi}{r \sin \theta}$$

y で偏微分すると

$$x \cos \phi \frac{\partial \phi}{\partial y} = \cos \phi + y(-\sin \phi) \frac{\partial \phi}{\partial y}$$

$$(r \sin \theta \cos \phi) \cos \phi + (r \sin \theta \sin \phi) \sin \phi \frac{\partial \phi}{\partial y} = \cos \phi$$

$$r \sin \theta \frac{\partial \phi}{\partial y} = \cos \phi$$

$$\frac{\partial \phi}{\partial y} = \frac{\cos \phi}{r \sin \theta}$$

z で偏微分すると

$$x \cos \phi \frac{\partial \phi}{\partial z} = -y \sin \phi \frac{\partial \phi}{\partial z}$$

$$\frac{\partial \phi}{\partial z} = 0$$

となる．さらに (1) (2) より

$$x^2 + y^2 = r^2 \sin^2 \theta$$

なので、両辺を x で偏微分すると

$$2x = \frac{\partial r^2}{\partial x} \sin^2 \theta + r^2 \frac{\partial \sin^2 \theta}{\partial x}$$

$$x = r \frac{\partial r}{\partial x} \sin^2 \theta + r^2 \sin \theta \cos \theta \frac{\partial \theta}{\partial x}$$

$\partial r / \partial x = \sin \theta \cos \phi$, $x = r \sin \theta \cos \phi$ を代入すると

$$r \sin \theta \cos \phi = r \sin \theta \cos \phi \sin^2 \theta + r^2 \sin \theta \cos \theta \frac{\partial \theta}{\partial x}$$

$$r \cos \theta \frac{\partial \theta}{\partial x} = \cos \phi (1 - \sin^2 \theta)$$

$$\frac{\partial \theta}{\partial x} = \frac{\cos \theta \cos \phi}{r}$$

両辺を y で偏微分すると

$$y = r \frac{\partial r}{\partial y} \sin^2 \theta + r^2 \sin \theta \cos \theta \frac{\partial \theta}{\partial y}$$

$\partial r / \partial y = \sin \theta \sin \phi$, $y = r \sin \theta \sin \phi$ を代入すると

$$r \sin \theta \sin \phi = r \sin \theta \sin \phi \sin^2 \theta + r^2 \sin \theta \cos \theta \frac{\partial \theta}{\partial y}$$

$$r \cos \theta \frac{\partial \theta}{\partial y} = \sin \phi (1 - \sin^2 \theta)$$

$$\frac{\partial \theta}{\partial y} = \frac{\cos \theta \sin \phi}{r}$$

両辺を z で偏微分すると

$$0 = r \frac{\partial r}{\partial z} \sin^2 \theta + r^2 \sin \theta \cos \theta \frac{\partial \theta}{\partial z}$$

$\partial r / \partial z = \cos \theta$ を代入すると

$$\frac{\partial \theta}{\partial z} = -\frac{\sin \theta}{r}$$

以上をまとめると

	x	y	z
r	$\sin \theta \cos \phi$	$\sin \theta \sin \phi$	$\cos \theta$
θ	$\cos \theta \cos \phi / r$	$\cos \theta \sin \phi / r$	$-\sin \theta / r$
ϕ	$-\sin \phi / r \sin \theta$	$\cos \phi / r \sin \theta$	0

となる.

ラプラシアンは

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

x について

$$\frac{\partial f}{\partial x} = \frac{\partial r}{\partial x} \frac{\partial f}{\partial r} + \frac{\partial \theta}{\partial x} \frac{\partial f}{\partial \theta} + \frac{\partial \phi}{\partial x} \frac{\partial f}{\partial \phi}$$

なので

$$\begin{aligned} \frac{\partial^2 f}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{\partial r}{\partial x} \frac{\partial f}{\partial r} + \frac{\partial \theta}{\partial x} \frac{\partial f}{\partial \theta} + \frac{\partial \phi}{\partial x} \frac{\partial f}{\partial \phi} \right) \\ &= \frac{\partial r}{\partial x} \frac{\partial}{\partial r} \left(\frac{\partial r}{\partial x} \frac{\partial f}{\partial r} + \frac{\partial \theta}{\partial x} \frac{\partial f}{\partial \theta} + \frac{\partial \phi}{\partial x} \frac{\partial f}{\partial \phi} \right) \\ &\quad + \frac{\partial \theta}{\partial x} \frac{\partial}{\partial \theta} \left(\frac{\partial r}{\partial x} \frac{\partial f}{\partial r} + \frac{\partial \theta}{\partial x} \frac{\partial f}{\partial \theta} + \frac{\partial \phi}{\partial x} \frac{\partial f}{\partial \phi} \right) \\ &\quad + \frac{\partial \phi}{\partial x} \frac{\partial}{\partial \phi} \left(\frac{\partial r}{\partial x} \frac{\partial f}{\partial r} + \frac{\partial \theta}{\partial x} \frac{\partial f}{\partial \theta} + \frac{\partial \phi}{\partial x} \frac{\partial f}{\partial \phi} \right) \end{aligned}$$

これに $\partial r / \partial x = \sin \theta \cos \phi$, $\partial \theta / \partial x = \cos \theta \cos \phi / r$, $\partial \phi / \partial x = -\sin \phi / r \sin \theta$ を代入すると

$$\begin{aligned}\frac{\partial r}{\partial x} \frac{\partial}{\partial r} \frac{\partial r}{\partial x} \frac{\partial f}{\partial r} &= \sin \theta \cos \phi \frac{\partial}{\partial r} \left(\sin \theta \cos \phi \frac{\partial f}{\partial r} \right) \\ &= \sin^2 \theta \cos^2 \phi \frac{\partial^2 f}{\partial r^2}\end{aligned}$$

$$\begin{aligned}\frac{\partial r}{\partial x} \frac{\partial}{\partial r} \frac{\partial \theta}{\partial x} \frac{\partial f}{\partial \theta} &= \sin \theta \cos \phi \frac{\partial}{\partial r} \left(\frac{\cos \theta \cos \phi}{r} \frac{\partial f}{\partial \theta} \right) \\ &= -\frac{\sin \theta \cos \theta \cos^2 \phi}{r^2} \frac{\partial f}{\partial \theta} + \frac{\sin \theta \cos \theta \cos^2 \phi}{r} \frac{\partial^2 f}{\partial r \partial \theta}\end{aligned}$$

$$\begin{aligned}\frac{\partial r}{\partial x} \frac{\partial}{\partial r} \frac{\partial \phi}{\partial x} \frac{\partial f}{\partial \phi} &= \sin \theta \cos \phi \frac{\partial}{\partial r} \left(-\frac{\sin \phi}{r \sin \theta} \frac{\partial f}{\partial \phi} \right) \\ &= \frac{\sin \phi \cos \phi}{r^2} \frac{\partial f}{\partial \phi} - \frac{\sin \phi \cos \phi}{r} \frac{\partial^2 f}{\partial r \partial \phi}\end{aligned}$$

$$\begin{aligned}\frac{\partial \theta}{\partial x} \frac{\partial}{\partial \theta} \frac{\partial r}{\partial x} \frac{\partial f}{\partial r} &= \frac{\cos \theta \cos \phi}{r} \frac{\partial}{\partial \theta} \left(\sin \theta \cos \phi \frac{\partial f}{\partial r} \right) \\ &= \frac{\cos^2 \theta \cos^2 \phi}{r} \frac{\partial f}{\partial r} + \frac{\sin \theta \cos \theta \cos^2 \phi}{r} \frac{\partial^2 f}{\partial \theta \partial r}\end{aligned}$$

$$\begin{aligned}\frac{\partial \theta}{\partial x} \frac{\partial}{\partial \theta} \frac{\partial \theta}{\partial x} \frac{\partial f}{\partial \theta} &= \frac{\cos \theta \cos \phi}{r} \frac{\partial}{\partial \theta} \left(\frac{\cos \theta \cos \phi}{r} \frac{\partial f}{\partial \theta} \right) \\ &= -\frac{\sin \theta \cos \theta \cos^2 \phi}{r^2} \frac{\partial f}{\partial \theta} + \frac{\cos^2 \theta \cos^2 \phi}{r^2} \frac{\partial^2 f}{\partial \theta^2}\end{aligned}$$

$$\begin{aligned}\frac{\partial \theta}{\partial x} \frac{\partial}{\partial \theta} \frac{\partial \phi}{\partial x} \frac{\partial f}{\partial \phi} &= \frac{\cos \theta \cos \phi}{r} \frac{\partial}{\partial \theta} \left(-\frac{\sin \phi}{r \sin \theta} \frac{\partial f}{\partial \phi} \right) \\ &= \frac{\cos^2 \theta \sin \phi \cos \phi}{r^2 \sin^2 \theta} \frac{\partial f}{\partial \phi} - \frac{\sin \phi \cos \theta \cos \phi}{r^2 \sin \theta} \frac{\partial^2 f}{\partial \theta \partial \phi}\end{aligned}$$

$$\begin{aligned}\frac{\partial \phi}{\partial x} \frac{\partial}{\partial \phi} \frac{\partial r}{\partial x} \frac{\partial f}{\partial r} &= -\frac{\sin \phi}{r \sin \theta} \frac{\partial}{\partial \phi} \left(\sin \theta \cos \phi \frac{\partial f}{\partial r} \right) \\ &= \frac{\sin^2 \phi}{r} \frac{\partial f}{\partial r} - \frac{\sin \phi \cos \phi}{r} \frac{\partial^2 f}{\partial \phi \partial r}\end{aligned}$$

$$\begin{aligned}\frac{\partial \phi}{\partial x} \frac{\partial}{\partial \phi} \frac{\partial \theta}{\partial x} \frac{\partial f}{\partial \theta} &= -\frac{\sin \phi}{r \sin \theta} \frac{\partial}{\partial \phi} \left(\frac{\cos \theta \cos \phi}{r} \frac{\partial f}{\partial \theta} \right) \\ &= \frac{\cos \theta \sin^2 \phi}{r^2 \sin \theta} \frac{\partial f}{\partial \theta} - \frac{\cos \theta \sin \phi \cos \phi}{r^2 \sin \theta} \frac{\partial^2 f}{\partial \phi \partial \theta}\end{aligned}$$

$$\begin{aligned}\frac{\partial \phi}{\partial x} \frac{\partial}{\partial \phi} \frac{\partial \phi}{\partial x} \frac{\partial f}{\partial \phi} &= -\frac{\sin \phi}{r \sin \theta} \frac{\partial}{\partial \phi} \left(-\frac{\sin \phi}{r \sin \theta} \frac{\partial f}{\partial \phi} \right) \\ &= \frac{\sin \phi \cos \phi}{r^2 \sin^2 \theta} \frac{\partial f}{\partial \phi} + \frac{\sin^2 \phi}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \phi^2}\end{aligned}$$

y について

$$\frac{\partial f}{\partial y} = \frac{\partial r}{\partial y} \frac{\partial f}{\partial r} + \frac{\partial \theta}{\partial y} \frac{\partial f}{\partial \theta} + \frac{\partial \phi}{\partial y} \frac{\partial f}{\partial \phi}$$

なので

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial r}{\partial y} \frac{\partial f}{\partial r} + \frac{\partial \theta}{\partial y} \frac{\partial f}{\partial \theta} + \frac{\partial \phi}{\partial y} \frac{\partial f}{\partial \phi} \right) = \frac{\partial r}{\partial y} \frac{\partial}{\partial r} \left(\frac{\partial r}{\partial y} \frac{\partial f}{\partial r} + \frac{\partial \theta}{\partial y} \frac{\partial f}{\partial \theta} + \frac{\partial \phi}{\partial y} \frac{\partial f}{\partial \phi} \right) + \frac{\partial \theta}{\partial y} \frac{\partial}{\partial \theta} \left(\frac{\partial r}{\partial y} \frac{\partial f}{\partial r} + \frac{\partial \theta}{\partial y} \frac{\partial f}{\partial \theta} + \frac{\partial \phi}{\partial y} \frac{\partial f}{\partial \phi} \right) + \frac{\partial \phi}{\partial y} \frac{\partial}{\partial \phi} \left(\frac{\partial r}{\partial y} \frac{\partial f}{\partial r} + \frac{\partial \theta}{\partial y} \frac{\partial f}{\partial \theta} + \frac{\partial \phi}{\partial y} \frac{\partial f}{\partial \phi} \right)$$

これに $\partial r / \partial y = \sin \theta \sin \phi$, $\partial \theta / \partial y = \cos \theta \sin \phi / r$, $\partial \phi / \partial y = \cos \phi / r \sin \theta$ を代入すると

$$\begin{aligned}\frac{\partial r}{\partial y} \frac{\partial}{\partial r} \frac{\partial r}{\partial y} \frac{\partial f}{\partial r} &= \sin \theta \sin \phi \frac{\partial}{\partial r} \left(\sin \theta \sin \phi \frac{\partial f}{\partial r} \right) \\ &= \sin^2 \theta \sin^2 \phi \frac{\partial^2 f}{\partial r^2}\end{aligned}$$

$$\begin{aligned}\frac{\partial r}{\partial y} \frac{\partial}{\partial r} \frac{\partial \theta}{\partial y} \frac{\partial f}{\partial \theta} &= \sin \theta \sin \phi \frac{\partial}{\partial r} \left(\frac{\cos \theta \sin \phi}{r} \frac{\partial f}{\partial \theta} \right) \\ &= -\frac{\sin \theta \sin^2 \phi \cos \theta}{r^2} \frac{\partial f}{\partial \theta} + \frac{\sin \theta \cos \theta \sin^2 \phi}{r} \frac{\partial^2 f}{\partial r \partial \theta}\end{aligned}$$

$$\begin{aligned}\frac{\partial r}{\partial y} \frac{\partial}{\partial r} \frac{\partial \phi}{\partial y} \frac{\partial f}{\partial \phi} &= \sin \theta \sin \phi \frac{\partial}{\partial r} \left(\frac{\cos \phi}{r \sin \theta} \frac{\partial f}{\partial \phi} \right) \\ &= -\frac{\sin \phi \cos \phi}{r^2} \frac{\partial f}{\partial \phi} + \frac{\sin \phi \cos \phi}{r} \frac{\partial^2 f}{\partial r \partial \phi}\end{aligned}$$

$$\begin{aligned}\frac{\partial \theta}{\partial y} \frac{\partial}{\partial \theta} \frac{\partial r}{\partial y} \frac{\partial f}{\partial r} &= \frac{\cos \theta \sin \phi}{r} \frac{\partial}{\partial \theta} \left(\sin \theta \sin \phi \frac{\partial f}{\partial r} \right) \\ &= \frac{\sin^2 \phi \cos^2 \theta}{r} \frac{\partial f}{\partial r} + \frac{\sin \theta \cos \theta \sin^2 \phi}{r} \frac{\partial^2 f}{\partial \theta \partial r}\end{aligned}$$

$$\begin{aligned}\frac{\partial \theta}{\partial y} \frac{\partial}{\partial \theta} \frac{\partial \theta}{\partial y} \frac{\partial f}{\partial \theta} &= \frac{\cos \theta \sin \phi}{r} \frac{\partial}{\partial \theta} \left(\frac{\cos \theta \sin \phi}{r} \frac{\partial f}{\partial \theta} \right) \\ &= -\frac{\sin \theta \cos \theta \sin^2 \phi}{r^2} \frac{\partial f}{\partial \theta} + \frac{\sin^2 \phi \cos^2 \theta}{r^2} \frac{\partial^2 f}{\partial \theta^2}\end{aligned}$$

$$\begin{aligned}\frac{\partial \theta}{\partial y} \frac{\partial}{\partial \theta} \frac{\partial \phi}{\partial y} \frac{\partial f}{\partial \phi} &= \frac{\cos \theta \sin \phi}{r} \frac{\partial}{\partial \theta} \left(\frac{\cos \phi}{r \sin \theta} \frac{\partial f}{\partial \phi} \right) \\ &= -\frac{\sin \phi \cos^2 \theta \cos \phi}{r^2 \sin^2 \theta} \frac{\partial f}{\partial \phi} + \frac{\cos \theta \sin \phi \cos \phi}{r^2 \sin \theta} \frac{\partial^2 f}{\partial \theta \partial \phi}\end{aligned}$$

$$\begin{aligned}\frac{\partial \phi}{\partial y} \frac{\partial}{\partial \phi} \frac{\partial r}{\partial y} \frac{\partial f}{\partial r} &= \frac{\cos \phi}{r \sin \theta} \frac{\partial}{\partial \phi} \left(\sin \theta \sin \phi \frac{\partial f}{\partial r} \right) \\ &= \frac{\cos^2 \phi}{r} \frac{\partial f}{\partial r} + \frac{\sin \phi \cos \phi}{r} \frac{\partial^2 f}{\partial \phi \partial r}\end{aligned}$$

$$\begin{aligned}\frac{\partial \phi}{\partial y} \frac{\partial}{\partial \phi} \frac{\partial \theta}{\partial y} \frac{\partial f}{\partial \theta} &= \frac{\cos \phi}{r \sin \theta} \frac{\partial}{\partial \phi} \left(\frac{\cos \theta \sin \phi}{r} \frac{\partial f}{\partial \theta} \right) \\ &= \frac{\cos \theta \cos^2 \phi}{r^2 \sin \theta} \frac{\partial f}{\partial \theta} + \frac{\cos \theta \sin \phi \cos \phi}{r^2 \sin \theta} \frac{\partial^2 f}{\partial \phi \partial \theta}\end{aligned}$$

$$\begin{aligned}\frac{\partial \phi}{\partial y} \frac{\partial}{\partial \phi} \frac{\partial \phi}{\partial y} \frac{\partial f}{\partial \phi} &= \frac{\cos \phi}{r \sin \theta} \frac{\partial}{\partial \phi} \left(\frac{\cos \phi}{r \sin \theta} \frac{\partial f}{\partial \phi} \right) \\ &= -\frac{\sin \phi \cos \phi}{r^2 \sin^2 \theta} \frac{\partial f}{\partial \phi} + \frac{\cos^2 \phi}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \phi^2}\end{aligned}$$

z について

$$\frac{\partial f}{\partial z} = \frac{\partial r}{\partial z} \frac{\partial f}{\partial r} + \frac{\partial \theta}{\partial z} \frac{\partial f}{\partial \theta} + \frac{\partial \phi}{\partial z} \frac{\partial f}{\partial \phi}$$

なので

$$\frac{\partial^2 f}{\partial z^2} = \frac{\partial}{\partial z} \left(\frac{\partial r}{\partial z} \frac{\partial f}{\partial r} + \frac{\partial \theta}{\partial z} \frac{\partial f}{\partial \theta} + \frac{\partial \phi}{\partial z} \frac{\partial f}{\partial \phi} \right) = \frac{\partial r}{\partial z} \frac{\partial}{\partial r} \left(\frac{\partial r}{\partial z} \frac{\partial f}{\partial r} + \frac{\partial \theta}{\partial z} \frac{\partial f}{\partial \theta} + \frac{\partial \phi}{\partial z} \frac{\partial f}{\partial \phi} \right) + \frac{\partial \theta}{\partial z} \frac{\partial}{\partial \theta} \left(\frac{\partial r}{\partial z} \frac{\partial f}{\partial r} + \frac{\partial \theta}{\partial z} \frac{\partial f}{\partial \theta} + \frac{\partial \phi}{\partial z} \frac{\partial f}{\partial \phi} \right) + \frac{\partial \phi}{\partial z} \frac{\partial}{\partial \phi} \left(\frac{\partial r}{\partial z} \frac{\partial f}{\partial r} + \frac{\partial \theta}{\partial z} \frac{\partial f}{\partial \theta} + \frac{\partial \phi}{\partial z} \frac{\partial f}{\partial \phi} \right)$$

これに $\partial r/\partial z = \cos \theta$, $\partial \theta/\partial z = -\sin \theta/r$, $\partial \phi/\partial z = 0$ を代入すると

$$\begin{aligned}\frac{\partial r}{\partial z} \frac{\partial}{\partial r} \frac{\partial r}{\partial z} \frac{\partial f}{\partial r} &= \cos \theta \frac{\partial}{\partial r} \left(\cos \theta \frac{\partial f}{\partial r} \right) \\ &= \cos^2 \theta \frac{\partial^2 f}{\partial r^2}\end{aligned}$$

$$\begin{aligned}\frac{\partial r}{\partial z} \frac{\partial}{\partial r} \frac{\partial \theta}{\partial z} \frac{\partial f}{\partial \theta} &= \cos \theta \frac{\partial}{\partial r} \left(-\frac{\sin \theta}{r} \frac{\partial f}{\partial \theta} \right) \\ &= \frac{\sin \theta \cos \theta}{r^2} \frac{\partial f}{\partial \theta} - \frac{\sin \theta \cos \theta}{r} \frac{\partial^2 f}{\partial r \partial \theta} \\ \frac{\partial r}{\partial z} \frac{\partial}{\partial r} \frac{\partial \phi}{\partial z} \frac{\partial f}{\partial \phi} &= 0\end{aligned}$$

$$\begin{aligned}\frac{\partial \theta}{\partial z} \frac{\partial}{\partial \theta} \frac{\partial r}{\partial z} \frac{\partial f}{\partial r} &= -\frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \left(\cos \theta \frac{\partial f}{\partial r} \right) \\ &= \frac{\sin^2 \theta}{r} \frac{\partial f}{\partial r} - \frac{\sin \theta \cos \theta}{r} \frac{\partial^2 f}{\partial \theta \partial r}\end{aligned}$$

$$\begin{aligned}\frac{\partial \theta}{\partial z} \frac{\partial}{\partial \theta} \frac{\partial \theta}{\partial z} \frac{\partial f}{\partial \theta} &= -\frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \left(-\frac{\sin \theta}{r} \frac{\partial f}{\partial \theta} \right) \\ &= \frac{\sin \theta \cos \theta}{r^2} \frac{\partial f}{\partial \theta} + \frac{\sin^2 \theta}{r^2} \frac{\partial^2 f}{\partial \theta^2} \\ \frac{\partial \theta}{\partial z} \frac{\partial}{\partial \theta} \frac{\partial \phi}{\partial z} \frac{\partial f}{\partial \phi} &= 0\end{aligned}$$

$$\frac{\partial \phi}{\partial z} \frac{\partial}{\partial \phi} \frac{\partial r}{\partial z} \frac{\partial f}{\partial r} = \frac{\partial \phi}{\partial z} \frac{\partial}{\partial \phi} \frac{\partial \theta}{\partial z} \frac{\partial f}{\partial \theta} = \frac{\partial \phi}{\partial z} \frac{\partial}{\partial r} \frac{\partial \phi}{\partial z} \frac{\partial f}{\partial \phi} = 0$$

$$\begin{aligned}
& \frac{\partial^2 f}{x^2} + \frac{\partial^2 f}{y^2} + \frac{\partial^2 f}{z^2} \\
&= \sin^2 \theta \cos^2 \phi \frac{\partial^2 f}{\partial r^2} \\
& - \frac{\sin \theta \cos \theta \cos^2 \phi}{r^2} \frac{\partial f}{\partial \theta} + \frac{\sin \theta \cos \theta \cos^2 \phi}{r} \frac{\partial^2 f}{\partial r \partial \theta} \\
& + \frac{\sin \phi \cos \phi}{r^2} \frac{\partial f}{\partial \phi} - \frac{\sin \phi \cos \phi}{r} \frac{\partial^2 f}{\partial r \partial \phi} \\
& + \frac{\cos^2 \theta \cos^2 \phi}{r} \frac{\partial f}{\partial r} + \frac{\sin \theta \cos \theta \cos^2 \phi}{r} \frac{\partial^2 f}{\partial \theta \partial r} \\
& - \frac{\sin \theta \cos \theta \cos^2 \phi}{r^2} \frac{\partial f}{\partial \theta} + \frac{\cos^2 \theta \cos^2 \phi}{r^2} \frac{\partial^2 f}{\partial \theta^2} \\
& + \frac{\cos^2 \theta \sin \phi \cos \phi}{r^2 \sin^2 \theta} \frac{\partial f}{\partial \phi} - \frac{\sin \phi \cos \theta \cos \phi}{r^2 \sin \theta} \frac{\partial^2 f}{\partial \theta \partial \phi} \\
& + \frac{\sin^2 \phi}{r} \frac{\partial f}{\partial r} - \frac{\sin \phi \cos \phi}{r} \frac{\partial^2 f}{\partial \phi \partial r} \\
& + \frac{\cos \theta \sin^2 \phi}{r^2 \sin \theta} \frac{\partial f}{\partial \theta} - \frac{\cos \theta \sin \phi \cos \phi}{r^2 \sin \theta} \frac{\partial^2 f}{\partial \phi \partial \theta} \\
& + \frac{\sin \phi \cos \phi}{r^2 \sin^2 \theta} \frac{\partial f}{\partial \phi} + \frac{\sin^2 \phi}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \phi^2} \\
& + \sin^2 \theta \sin^2 \phi \frac{\partial^2 f}{\partial r^2} \\
& - \frac{\sin \theta \sin^2 \phi \cos \theta}{r^2} \frac{\partial f}{\partial \theta} + \frac{\sin \theta \cos \theta \sin^2 \phi}{r} \frac{\partial^2 f}{\partial r \partial \theta} \\
& - \frac{\sin \phi \cos \phi}{r^2} \frac{\partial f}{\partial \phi} + \frac{\sin \phi \cos \phi}{r} \frac{\partial^2 f}{\partial r \partial \phi} \\
& + \frac{\sin^2 \phi \cos^2 \theta}{r} \frac{\partial f}{\partial r} + \frac{\sin \theta \cos \theta \sin^2 \phi}{r} \frac{\partial^2 f}{\partial \theta \partial r} \\
& - \frac{\sin \theta \cos \theta \sin^2 \phi}{r^2} \frac{\partial f}{\partial \theta} + \frac{\sin^2 \phi \cos^2 \theta}{r^2} \frac{\partial^2 f}{\partial \theta^2} \\
& - \frac{\sin \phi \cos^2 \theta \cos \phi}{r^2 \sin^2 \theta} \frac{\partial f}{\partial \phi} + \frac{\cos \theta \sin \phi \cos \phi}{r^2 \sin \theta} \frac{\partial^2 f}{\partial \theta \partial \phi} \\
& + \frac{\cos^2 \phi}{r} \frac{\partial f}{\partial r} + \frac{\sin \phi \cos \phi}{r} \frac{\partial^2 f}{\partial \phi \partial r} \\
& + \frac{\cos \theta \cos^2 \phi}{r^2 \sin \theta} \frac{\partial f}{\partial \theta} + \frac{\cos \theta \sin \phi \cos \phi}{r^2 \sin \theta} \frac{\partial^2 f}{\partial \phi \partial \theta} \\
& - \frac{\sin \phi \cos \phi}{r^2 \sin^2 \theta} \frac{\partial f}{\partial \phi} + \frac{\cos^2 \phi}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \phi^2} \\
& + \cos^2 \theta \frac{\partial^2 f}{\partial r^2} \\
& + \frac{\sin^2 \theta}{r} \frac{\partial f}{\partial \theta} - \frac{\sin \theta \cos \theta}{r} \frac{\partial^2 f}{\partial r \partial \theta} \\
& + \frac{\sin \theta \cos \theta}{r^2} \frac{\partial f}{\partial r} + \frac{\sin^2 \theta}{r^2} \frac{\partial^2 f}{\partial \theta \partial r} \\
& + \frac{\sin \theta \cos \theta}{r^2} \frac{\partial f}{\partial \theta} + \frac{\sin^2 \theta}{r^2} \frac{\partial^2 f}{\partial \theta^2}
\end{aligned}$$

$$= (\sin^2 \theta \cos^2 \phi + \sin^2 \theta \sin^2 \phi + \cos^2 \theta) \frac{\partial^2 f}{\partial r^2} + \left(+ \frac{\cos^2 \theta \cos^2 \phi}{r^2} + \frac{\sin^2 \phi \cos^2 \theta}{r^2} + \frac{\sin^2 \theta}{r^2} \right) \frac{\partial^2 f}{\partial \theta^2} + \left(\frac{\sin^2 \phi}{r^2 \sin^2 \theta} + \frac{\cos^2 \phi}{r^2 \sin^2 \theta} \right) \frac{\partial^2 f}{\partial \phi^2}$$

$$\begin{aligned} &= \frac{\partial^2 f}{\partial r^2} + \left(\frac{2}{r} \right) \frac{\partial f}{\partial r} + \frac{1}{r^2} \frac{\partial^2 f}{\partial \theta^2} + \left(\frac{\cos \theta}{r^2 \sin \theta} \right) \frac{\partial f}{\partial \theta} + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \phi^2} \\ &= \frac{1}{r^2} \left(r^2 \frac{\partial^2 f}{\partial r^2} + 2r \frac{\partial f}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \left(\sin \theta \frac{\partial^2 f}{\partial \theta^2} + \cos \theta \frac{\partial f}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \phi^2} \\ &= \frac{1}{r^2} \frac{\partial f}{\partial r} \left(r^2 \frac{\partial f}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial f}{\partial \theta} \left(\sin \theta \frac{\partial f}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \phi^2} \end{aligned}$$